Oracle service is implemented as a smart contract that provides a query function to access external sources. The oracle smart contract is imported into the calling smart contract and inherited. Then a query is used to access the oracle with the required data. The data requested is returned through a callback function because accessing data and verification may take some time. Figure 12.9 shows a simple class diagram relating a smart contract, oracle services, and an external data source.

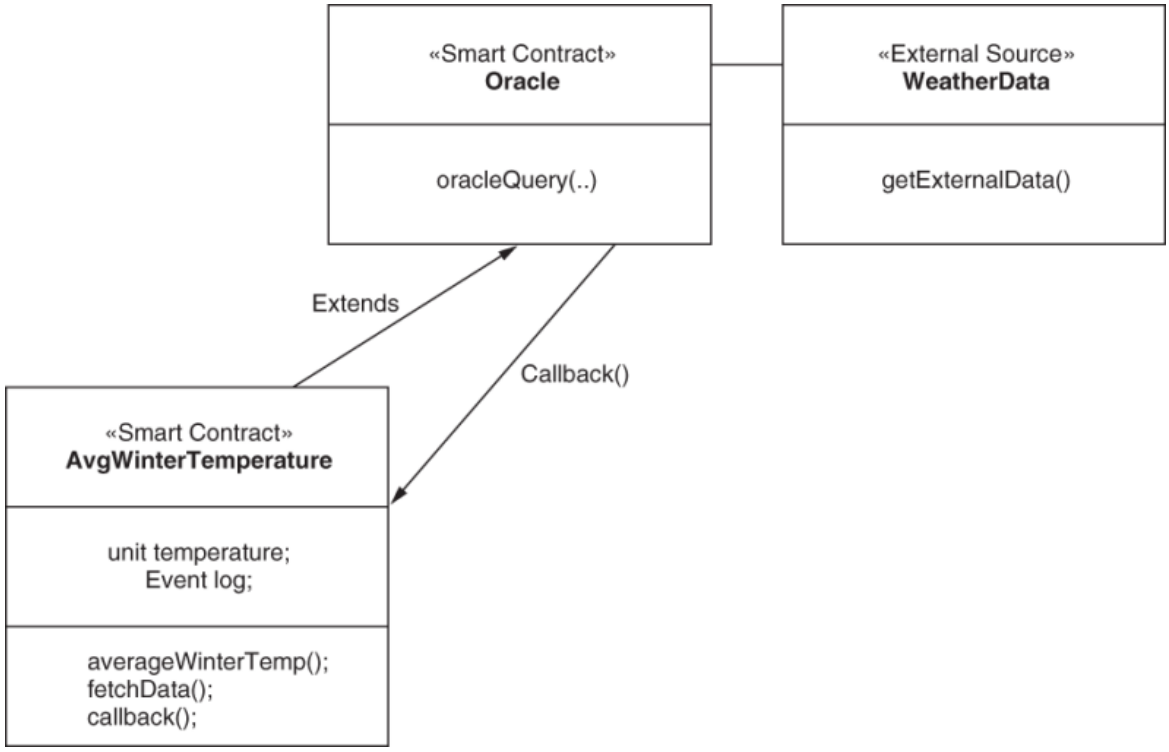


Figure 12.9 Accessing external data through oracle service

When the smart contract AvgWinterTemperature is deployed, it calls `fetchData()`, which in turn invokes the oracle service's query with the URL for the data source. It may take time to fetch the data, so a callback function is provided that can be invoked when the required data has been fetched. The oracle service accesses the external data source, authenticates the requested